

MySQL Advanced Practice Assignment: Tech Company Employee Performance & Sales Analytics

Assignment Overview

This assignment is designed to help students master **Subqueries, Joins, Window Functions, and CTEs** in MySQL using a realistic **Tech Company Employee Performance & Sales Analytics** dataset.

Database Schema

1. departments Table

```
CREATE TABLE departments (  
    department_id INT PRIMARY KEY AUTO_INCREMENT,  
    department_name VARCHAR(50) NOT NULL,  
    location VARCHAR(50) NOT NULL,  
    budget DECIMAL(15, 2) NOT NULL,  
    created_at DATE DEFAULT (CURRENT_DATE)  
);  
  
INSERT INTO departments (department_name, location, budget, created_at) VALUES  
(  
    ('Engineering', 'Bangalore, India', 5000000.00, '2022-01-15'),  
    ('Sales', 'Mumbai, India', 3000000.00, '2022-02-20'),  
    ('Marketing', 'Delhi, India', 2000000.00, '2022-03-10'),  
    ('HR', 'Hyderabad, India', 1500000.00, '2022-04-05'),  
    ('Finance', 'New York, USA', 4000000.00, '2022-05-12'),  
    ('Product', 'San Francisco, USA', 6000000.00, '2022-06-18'),  
    ('Support', 'Pune, India', 2500000.00, '2022-07-22'),  
    ('Operations', 'Chennai, India', 3500000.00, '2022-08-30');
```

2. employees Table

```
CREATE TABLE employees (  
    employee_id INT PRIMARY KEY AUTO_INCREMENT,  
    first_name VARCHAR(50) NOT NULL,  
    last_name VARCHAR(50) NOT NULL,  
    email VARCHAR(100) UNIQUE NOT NULL,  
    hire_date DATE NOT NULL,  
    salary DECIMAL(12, 2) NOT NULL,  
    department_id INT,
```

```
manager_id INT,  
job_title VARCHAR(50) NOT NULL,  
country VARCHAR(20) NOT NULL,  
FOREIGN KEY (department_id) REFERENCES departments(department_id) ON DELETE SET  
NULL,  
FOREIGN KEY (manager_id) REFERENCES employees(employee_id) ON DELETE SET NULL  
);
```

```
INSERT INTO employees (first_name, last_name, email, hire_date, salary,  
department_id, manager_id, job_title, country) VALUES  
( 'Rahul', 'Sharma', 'rahul.sharma@techcorp.com', '2022-01-10', 1200000.00, 1, NULL,  
'CTO', 'India'),  
( 'Priya', 'Patel', 'priya.patel@techcorp.com', '2022-02-15', 950000.00, 1, 1,  
'Senior Engineer', 'India'),  
( 'Amit', 'Kumar', 'amit.kumar@techcorp.com', '2022-03-20', 850000.00, 1, 1,  
'Engineer', 'India'),  
( 'Sneha', 'Mehra', 'sneha.mehra@techcorp.com', '2022-04-05', 800000.00, 1, 2,  
'Engineer', 'India'),  
( 'Rajesh', 'Gupta', 'rajesh.gupta@techcorp.com', '2022-05-12', 1500000.00, 2, NULL,  
'Sales Director', 'India'),  
( 'Ananya', 'Singh', 'ananya.singh@techcorp.com', '2022-06-18', 750000.00, 2, 5,  
'Sales Manager', 'India'),  
( 'Vikram', 'Jain', 'vikram.jain@techcorp.com', '2022-07-22', 600000.00, 2, 6, 'Sales  
Executive', 'India'),  
( 'Deepak', 'Chopra', 'deepak.chopra@techcorp.com', '2022-08-30', 550000.00, 2, 6,  
'Sales Executive', 'India'),  
( 'Lisa', 'Johnson', 'lisa.johnson@techcorp.com', '2022-09-10', 1300000.00, 5, NULL,  
'Finance Director', 'USA'),  
( 'Michael', 'Brown', 'michael.brown@techcorp.com', '2022-10-15', 900000.00, 5, 9,  
'Financial Analyst', 'USA'),  
( 'Sarah', 'Davis', 'sarah.davis@techcorp.com', '2022-11-20', 800000.00, 5, 9,  
'Accountant', 'USA'),  
( 'David', 'Wilson', 'david.wilson@techcorp.com', '2022-12-05', 1400000.00, 6, NULL,  
'Product Head', 'USA'),  
( 'Emily', 'Taylor', 'emily.taylor@techcorp.com', '2023-01-12', 1000000.00, 6, 12,  
'Product Manager', 'USA'),  
( 'James', 'Anderson', 'james.anderson@techcorp.com', '2023-02-18', 700000.00, 6, 13,  
'Product Designer', 'USA'),  
( 'Nina', 'Garcia', 'nina.garcia@techcorp.com', '2023-03-22', 650000.00, 6, 13, 'UX  
Researcher', 'USA'),  
( 'Arjun', 'Reddy', 'arjun.reddy@techcorp.com', '2023-04-10', 750000.00, 3, NULL,  
'Marketing Head', 'India'),  
( 'Ishita', 'Shah', 'ishita.shah@techcorp.com', '2023-05-15', 600000.00, 3, 16,  
'Marketing Specialist', 'India'),  
( 'Karan', 'Yadav', 'karan.yadav@techcorp.com', '2023-06-20', 550000.00, 3, 16,  
'Content Writer', 'India'),  
( 'Meera', 'Nair', 'meera.nair@techcorp.com', '2023-07-05', 1100000.00, 4, NULL, 'HR  
Director', 'India'),  
( 'Aditya', 'Verma', 'aditya.verma@techcorp.com', '2023-08-12', 500000.00, 4, 19, 'HR  
Executive', 'India'),  
( 'Rohit', 'Malhotra', 'rohit.malhotra@techcorp.com', '2023-09-22', 450000.00, 4, 19,  
'Recruiter', 'India'),  
( 'Sophia', 'Martinez', 'sophia.martinez@techcorp.com', '2023-10-10', 900000.00, 7,  
NULL, 'Support Head', 'India'),
```

```
('Ravi', 'Kumar', 'ravi.kumar@techcorp.com', '2023-11-15', 600000.00, 7, 22,
'Support Engineer', 'India'),
('Ankit', 'Sharma', 'ankit.sharma@techcorp.com', '2023-12-20', 550000.00, 7, 22,
'Support Engineer', 'India'),
('John', 'Robinson', 'john.robinson@techcorp.com', '2024-01-05', 1200000.00, 8,
NULL, 'Operations Director', 'India'),
('Priyanka', 'Choudhary', 'priyanka.choudhary@techcorp.com', '2024-02-12',
700000.00, 8, 25, 'Operations Manager', 'India');
```

3. projects Table

```
CREATE TABLE projects (
    project_id INT PRIMARY KEY AUTO_INCREMENT,
    project_name VARCHAR(100) NOT NULL,
    department_id INT,
    start_date DATE NOT NULL,
    end_date DATE,
    budget DECIMAL(15, 2) NOT NULL,
    status ENUM('Planned', 'In Progress', 'Completed', 'On Hold') DEFAULT 'Planned',
    FOREIGN KEY (department_id) REFERENCES departments(department_id) ON DELETE SET
NULL
);
```

```
INSERT INTO projects (project_name, department_id, start_date, end_date, budget,
status) VALUES
('AI Chatbot Development', 1, '2022-09-01', '2023-06-30', 2000000.00, 'Completed'),
('Cloud Migration', 1, '2023-01-15', '2023-12-31', 3000000.00, 'Completed'),
('Mobile App Redesign', 1, '2023-07-01', '2024-03-31', 1500000.00, 'In Progress'),
('Sales CRM Upgrade', 2, '2022-10-01', '2023-05-30', 1000000.00, 'Completed'),
('Enterprise Sales Push', 2, '2023-06-01', '2024-01-31', 2000000.00, 'In Progress'),
('Digital Marketing Campaign', 3, '2022-11-01', '2023-04-30', 800000.00,
'Completed'),
('Brand Awareness', 3, '2023-05-01', '2023-11-30', 1200000.00, 'Completed'),
('Financial Audit', 5, '2022-12-01', '2023-03-31', 500000.00, 'Completed'),
('Budget Planning', 5, '2023-04-01', '2023-09-30', 700000.00, 'Completed'),
('New Product Launch', 6, '2023-01-15', '2023-10-31', 2500000.00, 'In Progress'),
('Employee Engagement Survey', 4, '2023-02-01', '2023-06-30', 300000.00,
'Completed'),
('Customer Support Portal', 7, '2023-03-01', '2023-12-31', 900000.00, 'In
Progress'),
('Supply Chain Optimization', 8, '2023-04-01', '2024-01-31', 1500000.00, 'In
Progress');
```

4. project_assignments Table

```
CREATE TABLE project_assignments (
    assignment_id INT PRIMARY KEY AUTO_INCREMENT,
    employee_id INT NOT NULL,
    project_id INT NOT NULL,
    role VARCHAR(50) NOT NULL,
```

```

        start_date DATE NOT NULL,
        end_date DATE,
        hours_per_week INT DEFAULT 20,
        FOREIGN KEY (employee_id) REFERENCES employees(employee_id) ON DELETE CASCADE,
        FOREIGN KEY (project_id) REFERENCES projects(project_id) ON DELETE CASCADE,
        UNIQUE KEY (employee_id, project_id)
    );

```

```

INSERT INTO project_assignments (employee_id, project_id, role, start_date,
end_date, hours_per_week) VALUES

```

```

(2, 1, 'Lead Developer', '2022-09-01', '2023-06-30', 30),
(3, 1, 'Developer', '2022-09-01', '2023-06-30', 25),
(4, 1, 'Developer', '2022-09-01', '2023-06-30', 20),
(2, 2, 'Lead Developer', '2023-01-15', '2023-12-31', 35),
(3, 2, 'Developer', '2023-01-15', '2023-12-31', 30),
(4, 2, 'Developer', '2023-01-15', '2023-12-31', 25),
(3, 3, 'Lead Developer', '2023-07-01', '2024-03-31', 30),
(4, 3, 'Developer', '2023-07-01', '2024-03-31', 25),
(6, 4, 'Project Lead', '2022-10-01', '2023-05-30', 20),
(7, 4, 'Sales Executive', '2022-10-01', '2023-05-30', 15),
(8, 4, 'Sales Executive', '2022-10-01', '2023-05-30', 15),
(6, 5, 'Project Lead', '2023-06-01', '2024-01-31', 25),
(7, 5, 'Sales Executive', '2023-06-01', '2024-01-31', 20),
(8, 5, 'Sales Executive', '2023-06-01', '2024-01-31', 20),
(17, 6, 'Campaign Manager', '2022-11-01', '2023-04-30', 20),
(18, 6, 'Content Creator', '2022-11-01', '2023-04-30', 15),
(17, 7, 'Campaign Manager', '2023-05-01', '2023-11-30', 25),
(18, 7, 'Content Creator', '2023-05-01', '2023-11-30', 20),
(10, 8, 'Lead Auditor', '2022-12-01', '2023-03-31', 25),
(11, 8, 'Auditor', '2022-12-01', '2023-03-31', 20),
(10, 9, 'Lead Planner', '2023-04-01', '2023-09-30', 30),
(11, 9, 'Planner', '2023-04-01', '2023-09-30', 25),
(13, 10, 'Product Lead', '2023-01-15', '2023-10-31', 30),
(14, 10, 'Product Manager', '2023-01-15', '2023-10-31', 25),
(15, 10, 'Designer', '2023-01-15', '2023-10-31', 20),
(20, 11, 'Survey Lead', '2023-02-01', '2023-06-30', 20),
(21, 11, 'Surveyor', '2023-02-01', '2023-06-30', 15),
(23, 12, 'Portal Lead', '2023-03-01', '2023-12-31', 25),
(24, 12, 'Support Engineer', '2023-03-01', '2023-12-31', 20),
(25, 13, 'Operations Lead', '2023-04-01', '2024-01-31', 30);

```

5. sales_performance Table

```

CREATE TABLE sales_performance (
    performance_id INT PRIMARY KEY AUTO_INCREMENT,
    employee_id INT NOT NULL,
    month_year DATE NOT NULL,
    target DECIMAL(12, 2) NOT NULL,
    actual DECIMAL(12, 2),
    bonus DECIMAL(12, 2) DEFAULT 0.00,
    FOREIGN KEY (employee_id) REFERENCES employees(employee_id) ON DELETE CASCADE,
    UNIQUE KEY (employee_id, month_year)

```

```
);

INSERT INTO sales_performance (employee_id, month_year, target, actual, bonus)
VALUES
(5, '2022-10-01', 500000.00, 600000.00, 50000.00),
(5, '2022-11-01', 600000.00, 700000.00, 70000.00),
(5, '2022-12-01', 700000.00, 800000.00, 80000.00),
(6, '2022-10-01', 300000.00, 350000.00, 25000.00),
(6, '2022-11-01', 400000.00, 450000.00, 35000.00),
(6, '2022-12-01', 500000.00, 550000.00, 45000.00),
(7, '2022-10-01', 200000.00, 220000.00, 10000.00),
(7, '2022-11-01', 250000.00, 270000.00, 15000.00),
(7, '2022-12-01', 300000.00, 320000.00, 20000.00),
(8, '2022-10-01', 200000.00, 180000.00, 0.00),
(8, '2022-11-01', 250000.00, 260000.00, 5000.00),
(8, '2022-12-01', 300000.00, 310000.00, 10000.00),
(5, '2023-01-01', 650000.00, 750000.00, 75000.00),
(5, '2023-02-01', 700000.00, 800000.00, 80000.00),
(5, '2023-03-01', 750000.00, 850000.00, 85000.00),
(6, '2023-01-01', 450000.00, 500000.00, 40000.00),
(6, '2023-02-01', 500000.00, 550000.00, 45000.00),
(6, '2023-03-01', 550000.00, 600000.00, 50000.00),
(7, '2023-01-01', 300000.00, 330000.00, 20000.00),
(7, '2023-02-01', 350000.00, 380000.00, 25000.00),
(7, '2023-03-01', 400000.00, 420000.00, 30000.00),
(8, '2023-01-01', 300000.00, 290000.00, 0.00),
(8, '2023-02-01', 350000.00, 360000.00, 5000.00),
(8, '2023-03-01', 400000.00, 410000.00, 10000.00),
(5, '2023-04-01', 800000.00, 900000.00, 90000.00),
(5, '2023-05-01', 850000.00, 950000.00, 95000.00);
```

Practice Questions

Section A: Subqueries

Simple (1–3)

1. **Find all employees** who earn more than the average salary in their department.
2. **List the names and salaries** of employees who work in the 'Sales' department.
3. **Find all projects** that have a budget greater than the average project budget.

Medium (4–7)

1. **Find all employees** who have never been assigned to any project.
2. **List the names of employees** who have a salary higher than their manager's salary.
3. **Find all departments** where the average salary is greater than the company-wide average salary.

4. **List the names of employees** who have achieved their sales target in all recorded months.
-

Advanced (8–10)

1. **Find the top 3 highest-paid employees** in each department. Use a correlated subquery.
 2. **List the names of employees** who have worked on projects with a budget exceeding ₹2,000,000 and have also achieved at least 90% of their sales target in every month.
 3. **Find all employees** whose manager has never missed a sales target in any month.
-

Section B: Joins

Simple (1–3)

1. **List all employees** along with their department names and locations.
 2. **Find all projects** along with the names of the employees assigned to them.
 3. **List all sales performance records** with employee names, department names, and the month-year.
-

Medium (4–7)

1. **Find all employees** who have worked on more than 2 projects, along with the count of projects they've worked on.
 2. **List all departments** along with the total number of employees and the average salary in each department.
 3. **Find all employees** who have achieved their sales target in at least 3 different months, along with the months and actual sales.
 4. **List all projects** that are currently 'In Progress', along with the names of the employees working on them and their roles.
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Advanced (8–10)

1. **Find all employees** who have worked on projects in both the 'Engineering' and 'Sales' departments.
 2. **List all employees** who have a salary greater than the average salary of their department, along with their manager's name and salary.
 3. **Find all pairs of employees** who have worked together on at least 2 projects. Include employee names and the number of projects they've collaborated on.
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Section C: Window Functions

Simple (1–3)

1. **Rank employees** within each department by their salary in descending order.
 2. **Calculate the running total** of actual sales for each employee over the months.
 3. **Find the average salary** of employees in each department, along with each employee's salary and the difference from the department average.
-

Medium (4–7)

1. **Find the top 2 highest-paid employees** in each department, along with their rank and salary.
 2. **Calculate the percentage** of each employee's salary relative to the total salary of their department.
 3. **Find the moving average** of actual sales over a 3-month window for each employee.
 4. **List all employees** along with their salary, the average salary of their department, and the percentage difference between their salary and the department average.
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Advanced (8–10)

1. **Find employees** whose salary is in the top 25% of their department, along with their rank and salary.
 2. **Calculate the year-over-year growth** in actual sales for each employee, comparing each month to the same month in the previous year.
 3. **Find the top 3 employees** in each department based on their total bonus earned, along with their rank and total bonus.
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Section D: CTEs (Common Table Expressions)

Simple (1–3)

1. **Create a CTE** to find the average salary by department, then use it to list all employees with their salary and the average salary of their department.
 2. **Create a CTE** to find all employees who have achieved their sales target in all recorded months, then use it to list their names and total bonus earned.
 3. **Create a CTE** to find the total budget of all projects, then use it to list all projects with a budget greater than the average project budget.
-

Medium (4–7)

1. **Create a CTE** to find the top 3 highest-paid employees in each department, then use it to list their names, salaries, and department names.

2. **Create a CTE** to calculate the running total of actual sales for each employee, then use it to find employees whose running total exceeds ₹5,000,000.
 3. **Create a CTE** to find all employees who have worked on more than 2 projects, then use it to list their names, the number of projects, and their average hours per week.
 4. **Create a CTE** to find the average salary by job title, then use it to list all employees with their salary, job title, and the difference from the average salary for their job title.
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Advanced (8–10)

1. **Create a CTE** to find the top 2 employees in each department based on their total bonus earned, then use it to list their names, total bonus, and department names.
 2. **Create a CTE** to calculate the year-over-year growth in actual sales for each employee, then use it to find employees with positive growth in all recorded months.
 3. **Create a CTE** to find all pairs of employees who have worked together on at least 2 projects, then use it to list their names and the number of projects they've collaborated on.
-

Bonus Challenge

1. **Combine Subqueries, CTEs, and Window Functions** to find the top-performing employee in each department based on a weighted score (50% salary, 30% total bonus, 20% number of projects). List their names, departments, and weighted scores.
 2. **Write a query** to find all employees who have consistently outperformed their sales targets by at least 10% for 3 consecutive months, and also have a salary in the top 20% of their department. Use CTEs, Window Functions, and Subqueries.
 3. **Create a hierarchical report** showing the employee hierarchy (manager → employee) for the entire company, with each level indented. Use a recursive CTE to traverse the hierarchy.
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